

Fiscal 2025 R&D Subsidy (Overseas) Program Application Form

25/08/2025

I. Theme of R&D Application (Within 20 words)

Early Detection of Alzheimer's Disease and Mild Cognitive Impairment in Vietnamese older adults through speech analysis

II. Application Amount

Yen : 2,000,000

III. Applicant

Name	Ha Thi Thanh Huong				
Education level	PhD	Age	35	Gender	Female
Organization	International University-Vietnam National University Ho Chi Minh City	Position	Head of Department		
URL	https://hcmiu.edu.vn/en/				
Location	VNUHCM Township, Quarter 33, Linh Xuan Ward, Ho Chi Minh City				
E-mail	htthuong@hcmiu.edu.vn	Tel.	(+84) 938939445		
Affiliated academic societies	Member of IBRO Early Career Committee				
Summary of Applicant (Please state post-educational experience)	After graduating with a PhD degree from Stanford University, I decided to return to Vietnam in 2018. Upon coming back to Vietnam, I established the Brain Health Lab and have served as lecturer and department chair of Tissue Engineering and Regenerative Medicine at International University-Vietnam National University Ho Chi Minh City for over 6 years. My research focuses on addressing mental health issues and aging disorders to improve the quality of life. My team conducted an experiment that applies AI and a brain-computer interface system to support the rehabilitation of stroke patients. Additionally, we investigated AI potential in detecting brain tumors and migraines. Furthermore, I am interested in exploring the relationship between stress and emotions as indicated by changes in EEG signals, along with evaluating the effectiveness of techniques like alpha music and mindfulness-based stress reduction for alleviating stress. Besides, I am also interested in enhancing diagnosis and interventions for dementia at an early stage. In terms of the diagnosis aspect, I am leading the early diagnosis of Alzheimer's Disease, focusing on detecting p-Tau217 and cell-free RNA detection. For studies related to AI, I am developing an easy-to-use AI system to detect early signs of Alzheimer's disease in clinical settings or the biological characteristics of heterogeneity in the Mild Cognitive Impairment (MCI) population. When it comes to the intervention aspect, my research team developed BrainTrain for MCI patients to hinder their conversion to dementia and evaluate the efficacy of EEG signals in predicting post-intervention effectiveness. Our application has reached over 150 patients, providing accessible home-based dementia intervention for underserved communities. I am leading projects to build the first Vietnamese databases of neuroscience-related data, such as brain MRI and EEG signals to enhance AI models and improve diagnosis. Finally, my goal is to create innovative, highly accurate, and low-cost products that can have a tangible impact on the brain health and mental well-being of those in my community and beyond.				

IV. Recommender

I hereby recommend this candidate to TERUMO LIFE SCIENCE FOUNDATION's R&D Subsidies (Overseas) Program.

I am the head of the research institution to which the applicant belongs. I recommend only one application.

Name (Handwritten Signature)	Associate Professor Le Van Thang			
Organization	International University, Vietnam National University Ho Chi Minh City	Position	President	
E-mail	lvthang@hcmiu.edu.vn	Tel.	028 3724 4270	

Please keep this on the first page.

You may also insert figures and charts within the entry frames. However, please keep in mind that the total number of pages is within 6 pages.

V. Joint R&D Partners

Please list up to 3 names.

Name	Hoang Tien Trong Nghia		
Organization	175 Military Hospital	Position	Head of Neurology Department
Location	786 Nguyen Kiem Street, Hanh Thong Ward, Ho Chi Minh City		

Name	Ngo Thanh Hoan		
Organization	Florida Southern College	Position	Assistant Professor
Location	111 Lake Hollingsworth Drive, Lakeland, FL 33801, USA		

Name			
Organization		Position	
Location			

VI. Contact Information in Japan

Be sure to set up contact information in Japan. For example, researchers from research institutes in Japan, doctors in medical institutions in Japan etc.

Name	Taishiro Kishimoto		
Organization	Keio University School of Medicine	Position	Professor
URL	https://www.keio.ac.jp/en/		
Location	Mori JP Tower F7, 1-3-1, Azabudai, Minato-ku, Tokyo Postal Code: 106-0041		
E-mail	tkishimoto@keio.jp	Tel.	+81-3-5365-3219

VII. Subsidy Funds, or Supplemental Funds, from other Organizations with Regard to Related R&D

In the event an applicant is receiving subsidies from other funds with regard to this R&D theme, please state the name of funds, topic of research, period and amount for no more than three sources, starting with the latest.

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VIII. Abstract of Application Contents (about 160 words)

Please include the following points “the extent of contributions to society”, “originality” and “novelty” etc

By 2030, dementia will impact 78 million individuals globally. Alzheimer's disease (AD), which precedes mild cognitive impairment (MCI), affects 60–70% of people. Currently, individuals in rural regions of nations with limited resources lack the financial resources to access expensive brain imaging or specialized experts required for screening. Automated screening could be implemented with voice analysis, which is feasible solely in developed nations. Therefore, this project exploits the complex structure of the Vietnamese language system to optimize diagnosis by focusing on cognitive domains such as memory, attention, mathematics, and language, facilitating the identification of dementia in millions of Vietnamese speakers who are presently ignored by existing technology. The main outcome will be non-invasive, cost-effective screening technique that incorporates advanced Japanese speech analysis methods through international collaboration and also produces adapted speech tasks for Vietnamese individuals with limited education. This study introduces a novel early detection strategy that is culturally accessible for those with limited opportunities and serves as an initial framework for similar procedures in low- and middle-income countries globally.
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IX. Keywords

At least 1 word and no more than 5 words. If the target disease is clear, please include it in the keywords.

Alzheimer's Disease, MCI, translational research, Vietnamese speech and language impairment, Digital biomarkers

X. Purpose of the R&D

Please state what value this R&D will provide in actual medical settings or other areas, and expected contributions to society

Current cognitive screening is diagnosed on conventional neuropsychological assessments, which require trained clinicians and exhibit inter-rater variability. However, speech-based evaluation and analysis allow patients to independently perform standardized tasks, providing objective and consistent assessments. Therefore, the objective of this research is to establish a speech-based screening protocol that is culturally appropriate for the elderly Vietnamese population, such as sustained phonation, sentence reading, picture description, story recall, and verbal fluency tasks. Specifically, this study aims to evaluate the feasibility of deploying a speech-based assessment protocol for AD and MCI within Vietnamese healthcare environments and to identify specific acoustic and linguistic speech characteristics that effectively differentiate between AD, MCI, and healthy control groups during targeted speech tasks. Our approach addresses a critical accessibility challenge, since numerous individuals in rural and poor regions often lack awareness of early cognitive decline symptoms and postpone seeking medical assistance until significant damage arises. This method facilitates remote, economical screening using mobile and telemedicine platforms to identify cognitive decline early, thus alleviating the substantial healthcare burden and annual costs associated with late-stage dementia treatment. Additionally, our protocol establishes a basis for scalable telemedicine applications in Vietnam and supplies a replicable framework for creating speech methods across the Southeast Asia, thus facilitating equitable access to early cognitive screening in nations with insufficient clinical infrastructure.

XI. Summary of Previous Results from Applicant in Relation to This R&D Theme

This speech-based screening initiative utilizes the expertise in cognitive intervention technology and the cross-cultural validation experience of our diverse research team. We developed and executed the "BrainTrain" cognitive training program in partnership with the 175 Military Hospital to improve cognitive function in patients with cognitive impairments, demonstrating our ability to translate research into therapeutic tools. The validation of the MoCA-B and IQCODE cognitive assessments for Vietnamese elderly populations demonstrates our capability in culturally adapting international cognitive screening instruments. Furthermore, our research team conducted a literature review on acoustic and language-based analysis approaches in AD, which helped us establish our theoretical understanding of speech biomarkers and identify critical gaps in non-English language applications in AD. Additionally, we employed transfer learning techniques with BERT family models to develop machine learning models for Alzheimer's diagnosis. These models were applied to English speech datasets from the DementiaBank Pitt Corpus, highlighting the critical need for language-specific protocols and the challenges relating to cross-linguistic speech analysis.

XII. Details of Subsidy Purpose

Please provide an approximate breakdown of the "II. Application Amount" by Japanese Yen, in line with experiment equipment, experiment tools, test reagents, and other items.

Item	Unit	Price (JPY)	Description	Total (JPY)
A. Equipment				134,611
Zoom Handheld Recorder	H6 2	58,205	Professional 6-track digital audio recorder: 24-bit/96 kHz recording, interchangeable microphone capsules (X/Y, Mid-Side), 4 XLR/TRS inputs with phantom power, direct SD recording up to 128 GB, 20+ hour battery life.	116,410
Audio accessories (cables, memory cards)		5,598	Cables, high-speed SD cards (32GB Class 10)	5,598
Data storage with Google One for 12 months	1	12,603	Secure, backed-up server space for storing sensitive patient data	12,603
B. Research material and participant support				485,667

Cognitive assessment materials, printing, and documentation		11,197	This fee will cover the costs of informed consent documents and the assessment protocol. The materials include instruction cards and visual aids. Professional printing ensures standardized stimulus. The presentation is critical for ensuring reliable data collection from 160 participants.	11,197
Participant support	170	2,791	This fee compensates each participant with 1,675 JPY, and the remaining fee of 1,116 JPY covers the hospital's medical expenses for 160 participants and 10 pilot participants.	474,470
C. Clinical Assessment, medical consultation, and technician support				804,100
Clinical Assessment and Medical Supervision	170	3,910	Expert Clinical Services (3,910 JPY per participant): neurological assessment, DSM-5 criteria application, medical supervision, clinical interpretation, and participant safety monitoring for 160 participants and 10 pilot participants.	664,700
Technician support	170	820	Each technical personnel member is paid 820 JPY per participant for data collection, equipment operation, participant scheduling, preliminary processing, and transcribing audio into text over six months.	139,400
D. Dissemination:				279,707
Conference presentations, publication fees, and patent fees		279,707	Publication fees include support for open-access publications (about 83,728 JPY) in neuroscience and medical journals to maximize research impact and establish international research in the dementia research community. The remainder (about 195,979 JPY) will be allocated for conference attendance.	279,707
E. Travel				295,915
Master's Student Training Exchange to Japan (2 weeks)	1	295,915	The funding will cover the travel fee to Japan for training in advanced speech analysis methodology at Prof. Dr. Kishimoto's laboratory, accommodation, and living expenses during the research exchange.	295,915
Total (JPY)				2,000,000

XIII. R&D Implementation Plan

Please state the experiment and testing process, and their schedules.

Please state the division of roles in the event there is to be a partner jointly conducting the R&D.

The study will be split into two phases:

Phase 1: Developing the speech protocol and assessing feasibility.

Phase 1.1 (2 months): Protocol finalization and ethics approval

1. Determine and establish the core content framework of the speech task for our protocol.
2. Optimize our protocol based on feedback and recommendations from specialists
3. Obtain research ethics approval from 175 Military Hospital

Role Division: Dr. Nghia and I will develop speech tasks like sustained phonation, sentence reading, picture description, story recall (both immediate and delayed), and verbal fluency (semantic and phonemic), and set a data collection timeline. Dr. Nghia will provide clinical knowledge for protocol validation, verify hospital ethics compliance, and set up our hospital experience to collect quality data.

Phase 1.2 (1 month): Pilot study execution

1. Recruit 15 participants (5 AD, 5 MCI) to evaluate three key criteria:
 - 1.1. **Protocol Feasibility:** protocol compliance, audio quality, and evaluating tasks in the protocol.
 - 1.2. **Resource Feasibility:** Evaluate recording equipment requirements and participant completion time.
 - 1.3. **Management Feasibility:** Assess research personnel capacity and coordination requirements.
2. Optimize speech tasks in our protocol by incorporating pilot feedback from clinicians and technicians and utilizing a 4-point scale scoring system (1-4) that is based on three validation criteria:
 - 2.1. **Clarity and Appropriateness:** Are the images and narratives clear, comprehensible, and suitable for elderly participants?
 - 2.2. **Task Instructions:** Are the protocol task instructions sufficiently simple and straightforward?

2.3. Scoring Criteria: Are the assessment criteria clear, reasonable, and practically applicable?

Validation Threshold: Questions receiving scores of 3 or 4 from more than 80% of expert evaluators across all criteria (Yusoff, 2019¹) will be approved for inclusion in the main study protocol.

3. Refine data collection procedures

Role Division: The clinical team at 175 Military Hospital will help assess each participant and classify them into the AD or MCI group for the pilot study testing. The clinical team also supervises the pilot data collection process. Finally, my research team will obtain feedback from clinicians and technicians on our questions and scores to enhance our procedure and speech tasks.

Phase 2: Main collection.

Phase 2.1 (6 months): Participant Recruitment and Clinical Assessment

- Recruit 150 participants (aged over 55) from the Military Hospital 175 outpatient clinics
- Inform all participants about the study objectives and obtain informed consent
- Clinical neurologists evaluate participants according to inclusion/exclusion criteria

Inclusive Criteria	Exclusive Criteria
- Participants who agreed to participate in the study. - 55 years of age or older. - MCI patients diagnosed according to DSM-5 criteria (DSM5-MCI) - AD patients diagnosed according to DSM-5 criteria (DSM5-AD)	- Participants who do not consent to participate in the study - Treatments that include cholinesterase inhibitors or other medications that may affect cognitive abilities. - Diseases that affect the central nervous system (multiple sclerosis, Parkinson's disease, stroke, etc.).

- Distribute diagnosed participants into three groups: 50 AD, 50 MCI, 50 Healthy Controls (HC)

Role Division: The neurology team at 175 Military Hospital will conduct comprehensive clinical assessments using DSM-5 criteria to diagnose and classify participants into AD, MCI, and HC groups.

Phase 2.2 (6 months): Speech Data Collection

- Conduct optimized speech tasks with all 150 participants.
- Record participant speech performance with cognitive tasks including sustained phonation, sentence reading, picture description, story recall (both immediate and delayed), and verbal fluency (semantic and phonemic).
- Implement quality control measures for audio data integrity.

Role Division: Our trained technicians will conduct the audio recording sessions and administer the speech protocol tasks. Assistant Professor Ngo Thanh Hoan will provide remote consultation on data quality assurance. In this phase, I will manage the collection process and address any unforeseen events that can occur.

Phase 2.3 (2 months): Feature Extraction and Analysis

- Extract acoustic and linguistic features from all audio recordings.
- Conduct a statistical analysis to validate extracted speech features for diagnostic accuracy
- This phase will involve close collaboration with Prof. Dr. Taishiro Kishimoto. A master's student from our team will join Prof. Dr. Kishimoto's laboratory for intensive training in speech analysis methodologies and feature extraction techniques. This exchange will provide hands-on training in advanced feature extraction methods at Prof. Dr. Kishimoto's lab, direct technical consultation on acoustic and linguistic feature selection, and facilitate transferring specialized knowledge to our research team in Vietnam.

Role Division: Prof. Dr. Kishimoto's team will teach state-of-the-art speech analysis techniques. This will ensure proper implementation of feature extraction methods and establish a strong foundation for ongoing collaboration. Assistant Professor Ngo Thanh Hoan will contribute his expertise in statistical validation of the speech features. My master's student and I will lead the data analysis process, incorporating the knowledge and techniques learned during the Japan exchange.

Phase 2.4 (2 months): Results analysis and dissemination

- Complete analysis and interpretation
- Writing a report and preparing a manuscript for submission to international journals
- Develop clinical implementation guidelines for the validated protocol

Role Division: My master's student and I will lead the manuscript preparation with contributions from all collaborators. Prof. Dr. Kishimoto will discuss clinical implications and international perspectives, while Assistant Professor Ngo Thanh Hoan will focus on the technical methodology sections.

XIV. Overview of R&D Trends and Advantages with Regard to This R&D Theme

Please state the advantages of your research over current diagnostics, medical procedures, or competing medical technologies and also include how much the advantages is compared to them.

¹ [78] Department of Medical Education, School of Medical Sciences, Universiti Sains Malaysia, MALAYSIA and M. S. B. Yusoff, "ABC of Content Validation and Content Validity Index Calculation," EIMJ, vol. 11, no. 2, pp. 49-54, June 2019.

In Vietnam, the validated Mini-Mental State Examination (MMSE) and Montreal Cognitive Assessment (MoCA) are the primary standardized neuropsychological tests that are currently used for cognitive screening [1, 2]. These assessments are low-cost and widely used in clinical practice, but they have several drawbacks, including inter-rater variability, trained healthcare professionals, and rural Vietnam's elderly low-educated population. Advanced neuroimaging techniques like MRI and PET scans are accurate but cost approximately 13,955-139,547 JPY per assessment and require radiological expertise that is scarce in rural areas, where elderly people have the highest MCI prevalence. Speech-based cognitive assessment has revolutionized early detection, with numerous studies showing that acoustic and linguistic changes in speech [3, 4]. Most speech-based diagnosis approaches are for developed countries, which is impossible for older Vietnamese, especially in rural and isolated locations, who have poor or low education due to previous educational restrictions and socioeconomic issues. Effective modifications for Mandarin Chinese, Japanese, and Spanish show that language-specific protocols can preserve diagnostic accuracy while resolving cultural and linguistic differences [5]. Vietnamese has six tones and unique phonetic and prosodic characteristics that differ from non-tonal languages, requiring language-specific modeling to capture tone variations and specific speech rhythm patterns that may be affected by cognitive decline [6]. Our Vietnamese speech-based screening methodology tackles these limitations by using cost-effective assessment, simple recording equipment available in community health clinics, and appropriate tasks for low-education individuals. This approach offers quantitative, objective measurement compared to subjective clinical observations, remote deployment and scalability across Vietnam's diverse healthcare infrastructure, and early detection before clinical dementia onset when interventions may be most effective.

1. M. Do et al., "Validation of MoCA Test in Vietnamese Language for Cognitive Impairment Screening," JoGHNP, p. e2022008, May 2022.
2. N. K. Quốc and V. A. Nhi, "Khảo sát Thang điểm MMSE trên Người Việt Bình thường," TCYHTPHCM, vol. 10, no. 1, pp. 237–237, 2006.
3. C. De Looze et al., "Changes in Speech Chunking in Reading Aloud is a Marker of Mild Cognitive Impairment and Mild-to-Moderate Alzheimer's Disease," Curr Alzheimer Res, vol. 15, no. 9, pp. 828–847, 2018.
4. F. Haider, S. De La Fuente, and S. Luz, "An assessment of paralinguistic acoustic features for detection of Alzheimer's dementia in spontaneous speech," IEEE Journal of Selected Topics in Signal Processing, vol. 14, no. 2, pp. 272–281, 2019.
5. García, A. M., de Leon, J., Tee, B. L., Blasi, D. E., & Gorno-Tempini, M. L. (2023). Speech and language markers of neurodegeneration: a call for global equity. Brain: a journal of neurology, 146(12), 4870–4879.
6. K. D. Le, "Development of a new aphasia test for Vietnamese people (Vietnamese aphasia test)," thesis, Open Research Newcastle, 2025. Accessed: July 17, 2025.

XV. Representative Research Papers by the Individual Applying for This Subsidy that Relate to the Theme of This R&D

Please list in the order of Author, Article title, Journal title, Date of publication, Volume number, Issue number, Page number(s) within approx. the past 5 years.

Please designate the research paper which you feel is most representative.

It is not necessary to attach the research papers themselves.

1. Thanh Truc-Quynh Nguyen, Chau Bao Dinh Hoang, Minh Dung Hoang Le, Ngoc Tram Anh V, Halle Quang, Christopher Minh Nguyen, Claire Goodman, George M Savva, Van Luan Pham, Trung Thu Tran, Van Toi Vo, and **Thi Thanh Huong Ha**; Assessing cognitive decline in Vietnamese older adults using the Montreal Cognitive Assessment-Basic (MoCA-B) and Informant Questionnaire on Cognitive Decline in the Elderly (IQCODE) during the COVID-19 pandemic: a feasibility study, The Clinical Neuropsychologist, 2023, Volume 37, Issue 5, Pages 1043–1061.
2. Thanh, N.N.T., **Thanh, H.H.T.**, Thanh, H.P., Thi, L.N., Trong, N.H.T., Hoai, T.T.T.; Recent Approaches on Acoustic and Language-Based Analysis in Alzheimer's Disease: A Literature Review, 10th International Conference on the Development of Biomedical Engineering in Vietnam, BME 2024, IFMBE Proceedings, 2025, Volume 123, Pages 37.
3. Nguyen, X.K., Ngo, T.A., Nguyen, T., Pham, P., **Ha, H.**, Ngo, L.; A Comparative Study of Transfer Learning Techniques with BERT Family Models in Early Diagnosis of Alzheimer's Disease, 10th International Conference on the Development of Biomedical Engineering in Vietnam, BME 2024, IFMBE Proceedings, 2025, Volume 122, Pages 26.
4. Phan T. N. M., Nguyen H. T., Huynh T. N. M., Nguyen T. H., Tran T. N. Y., and **Ha H. T. T.**, Development and Commercialization of a Brain Training App Targeting the Vietnamese Elderly, International Conference on the Development of Biomedical Engineering in Vietnam, December 2022, Pages 719-737.
5. Duong H., Nguyen T., Pham D., Duong K., Mieu L., Nguyen H., **Ha H. T. T.**, and Ngo L., HosRev: A Comprehensive Aspect-Category Sentiment Analysis Dataset on Hospital Reviews in Vietnam, 2023 1st International Conference on Health Science and Technology (ICHST), December 2023, Pages 1-6.

Please keep this within 6 pages.